ThemeCloner: Holdings-Free Discovery of Thematic Exposure via Latent Factor Projection

Working Draft

Abstract

Thematic investing groups companies by exposure to a structural economic driver — artificial intelligence, clean energy, cybersecurity — rather than by traditional sector classification. Yet the baskets that define a theme are typically hand-built, large-cap, and backward-looking, and it is rarely clear in advance which themes are economically coherent enough to trade. We present ThemeCloner, a holdings-free engine that extracts a theme's statistical signature from the return co-movement of thematic exchange-traded funds (ETFs) in a broad large- and mid-cap universe, then projects that signature onto a disjoint small-cap universe to discover stocks with genuine thematic exposure that no classification or ETF holding has labeled. Using principal component analysis on factor-residualized, standardized returns, our central result is a diagnostic: a theme produces clean, investable baskets if and only if it uniquely owns a latent factor. Concentration alone is insufficient — a theme can be concentrated on a factor it shares with others, yielding baskets that look clean but are contaminated — so it is distinctness of factor ownership, not coherence per se, that determines tradeability. This single criterion also predicts which themes contaminate one another and, in a direct momentum battery, which themes' discovered signal survives controls for stock-level momentum, supplying a precise, mechanism-based answer to the recurring claim that thematic investing is merely momentum. To validate that the diagnostic measures genuine structure rather than rediscovering taxonomy, we calibrate it against GICS sectors and subsectors as control labels. The calibration yields a striking result: sectors do not define the coherence ceiling. Coherence is regime-dependent rather than taxonomic — in the 2018-2026 sample, several cross-sector themes are more distinctly-owned latent factors than the GICS sectors they span, because a structural driver has fractured those sectors internally. The work offers both a portable discovery tool and a pre-trade test for when a theme is economically real rather than a marketing label.

1. Introduction

Thematic investing has become one of the most prominent organizing ideas in modern asset management. Investors increasingly want exposure to structural shifts — artificial intelligence, the energy transition, cybersecurity, automation — that cut across the boundaries of traditional sector taxonomies. The industry packages these ideas into thematic ETFs, which group companies by shared exposure to a trend rather than by their Global Industry Classification Standard (GICS) sector. The premise is that the relevant axis of co-movement is the theme, not the sector.

Three difficulties undermine this premise in practice. First, thematic baskets are backward-looking: by the time an index provider names a theme and an ETF lists, the defining companies have often been moving together for years, and the basket is anchored to the large-cap names that already led. Second, the baskets are hand-built from holdings data and fundamental classification, which makes them costly to construct, opaque, and blind to the small-cap companies that may carry the theme's purest exposure. Third, there is no principled way to know in advance which themes are economically coherent enough to trade: some themes correspond to genuine, distinct patterns of co-movement, while others are taxonomic labels imposed on companies that do not actually move together. A recurring skeptical claim — voiced informally across the industry but never, to our knowledge, tested directly — sharpens this third difficulty: that thematic investing is simply momentum in disguise, earning its returns because its members recently rose rather than because they share an economic driver.

This paper addresses all three. We ask whether a theme's signature can be recovered from return co-movement alone, without holdings data, and whether that signature can be projected onto an unexplored universe to discover thematically-exposed stocks that no classification has labeled. We ask, given such a method, what distinguishes a theme that yields clean, investable baskets from one that does not. And we ask whether the discovered signal is genuinely distinct from momentum.

Our approach, ThemeCloner, treats thematic ETF returns as labels for a latent factor rather than as a list of constituents. We residualize a broad large- and mid-cap universe against standard risk factors, extract latent factors from the residual co-movement via principal component analysis, and identify each theme's fingerprint by regressing its ETFs onto those factors. Projecting a disjoint small-cap universe into the same factor space and scoring by similarity to each fingerprint yields candidate baskets. The method never uses ETF holdings; the ETF price series is the theme definition.

Our central finding is a diagnostic: a theme produces clean, investable baskets if and only if it uniquely owns a latent factor. Coherence — how concentrated a theme's fingerprint is — is necessary but not sufficient, because a theme can be concentrated on a factor it shares with other themes, producing baskets that look clean but are contaminated. Distinctiveness — whether the theme owns its dominant factor — is the operative criterion. This single mechanism turns out to explain three otherwise-separate phenomena: which themes yield good baskets, why commodity-linked themes contaminate one another, and which themes survive a momentum control. It is, in effect, a pre-trade test of whether a theme is economically real, computable before any basket is built.

To establish that this diagnostic measures genuine structure rather than merely rediscovering sector taxonomy, we calibrate it against GICS sectors and subsectors as control labels, expecting them to define the coherence ceiling — a sector should own a factor almost by definition. The calibration both validates the diagnostic and yields a striking result in its own right: sectors do not sit at the top. Coherence proves to be regime-dependent rather than a fixed property of taxonomy — in our 2018-2026 sample, several cross-sector themes are more distinctly-owned factors than the broad GICS sectors they draw from, because the dominant structural driver of the period, capital expenditure on artificial intelligence, has split those sectors internally. That a cross-sector theme can be a tighter covariance cluster than the sector it spans is the strongest possible evidence that the diagnostic captures real structure GICS misses; we develop it as a validation of the method rather than as a standalone claim about thematic investing's legitimacy.

We make four contributions. We present a holdings-free, cross-universe engine for thematic stock discovery. We introduce a coherence-and-distinctiveness diagnostic that determines, before any basket is built, whether a theme is a genuine covariance cluster or a taxonomic label — the paper's central contribution. We provide direct evidence on the thematic-versus-momentum question, showing the answer is conditional on factor ownership. And, calibrating the diagnostic against GICS controls, we document regime-dependent coherence, which validates that the method resolves structure finer than sector classification.

2. Related Work

2.1 Latent Factor Models of Returns

Our factor-extraction step builds on the literature that estimates latent asset-pricing factors statistically rather than specifying them a priori. The risk-premium PCA of Lettau and Pelger generalizes principal component analysis by penalizing pricing error, surfacing weak factors with high Sharpe ratios that ordinary PCA can miss. We adopt the residual-covariance machinery of this tradition but, because our object is theme discovery rather than stochastic-discount-factor construction, we show the risk-premium penalty is inactive on mean-zero residuals and reduces to ordinary PCA. Our themes are, in the language of this literature, weak factors — they explain a modest share of residual variance — which is precisely why a covariance-structured method is required to recover them.

2.2 Text-Based and Classification-Free Company Grouping

A parallel literature derives company relationships from sources other than returns. Text-based network industries built from regulatory filings produce peer groups that differ from fixed classifications such as SIC or GICS, and recent work learns company embeddings from disclosures that recover industry and similarity structure. A complementary approach to ours derives themes from disclosure language rather than return co-movement and tests whether language-based theme membership anticipates later ETF inclusion (Amini et al., working paper). These approaches share our goal of moving beyond fixed taxonomy but operate on text; ThemeCloner operates on the covariance of returns, and the two are orthogonal signals on the same underlying question of what truly groups firms.

2.3 Thematic Investing and Its Coherence

The closest work to ours examines thematic investing from a risk-based perspective, defining a theme through a factor model and testing whether thematic baskets capture real, coherent common risk. That work tests the coherence of given baskets; we differ in objective and direction — we discover basket members across disjoint universes, and we use coherence not only to validate but to predict which themes are tradeable at all. Related practitioner work argues that companies within a theme should correlate with the theme's macro drivers while decorrelating from adjacent sectors, which is the fundamental-data counterpart to the statistical separation we measure.

2.4 Momentum and the Thematic Critique

The claim that thematic investing is repackaged momentum has not, to our knowledge, been tested directly. The most relevant evidence is the result that momentum is not a standalone factor but the time-series autocorrelation of other factor returns, and that factor momentum subsumes much of individual-stock momentum. This frames our distinction precisely: momentum is a temporal property of factor returns, whereas a theme is a contemporaneous cross-sectional covariance structure. The two are different mathematical objects, and we test their separation empirically rather than asserting it.

3. Methodology

ThemeCloner extracts the statistical signature of an investment theme from the return co-movement of thematic exchange-traded funds (ETFs), then projects that signature onto a separate, unexplored universe of stocks to identify names with genuine thematic exposure. The method is holdings-free by construction: ETF price series are used as theme labels, and no constituent data is required at any stage. This section describes the three-universe architecture, the residualization and factor-extraction steps, and the diagnostics that determine when a theme is recoverable.

3.1 Three-Universe Architecture

The pipeline separates three distinct roles that are often conflated in thematic analysis:

- Theme-definition universe (thematic ETFs). Each theme is represented by two to three US-listed thematic ETFs. Their returns define the theme and serve as the out-of-sample validation benchmark. Holdings are never used.
- Covariance universe (large and mid cap). The cross-section on which latent factors are estimated. We use US-listed stocks above a USD 1 billion market-capitalization floor (approximately 2,500 names), restricted to a single timezone to avoid foreign-exchange and price-snap timing artifacts.
- Target universe (small cap). The universe in which candidates are sought: NASDAQ and NYSE names below the covariance-universe floor, metadata-filtered, approximately 2,100 names. This is where the method hunts for previously unlabeled thematic exposure.

Keeping these universes distinct is essential. The covariance universe must be broad and clean enough that themes appear as genuine latent factors; the target universe must be disjoint so that discovered names are genuinely new rather than re-identified ETF constituents.

3.2 Residualization

Raw returns are dominated by market, size, value, and momentum exposures. Running factor extraction on raw returns causes any high-beta or high-momentum stock to load on every theme, producing pervasive contamination. We therefore residualize all return series against the Fama-French five factors (optionally augmented with the momentum factor and with commodity-sector controls; see Section 4) before factor extraction. The residual is the component of each stock's return that standard risk factors cannot explain.

Residuals are then winsorized at the 1st and 99th percentiles and standardized to unit volatility. This converts the subsequent decomposition from a covariance to a correlation analysis, preventing any single high-volatility name from dominating the extracted factors. This standardization step proved decisive: without it, factors were driven by individual volatile micro-caps; with it, every factor became an interpretable sector or style axis.

3.3 Factor Extraction

We extract K = 15 latent factors from the standardized residuals of the covariance universe. We adopt principal component analysis (PCA) on the residuals rather than risk-premium PCA (RP-PCA). The reason is structural: RP-PCA augments the PCA objective with a penalty on the cross-sectional mean return, but residualization renders the residuals mean-zero by construction, so the penalty term vanishes and RP-PCA reduces exactly to PCA. We confirmed empirically that theme detectability is invariant to the RP-PCA penalty weight across a range spanning two orders of magnitude. Because theme discovery is fundamentally a question about co-movement structure rather than about constructing a high-Sharpe stochastic discount factor, PCA on standardized residuals is the appropriate tool. The choice of K = 15 is justified by a factor-count sweep (Appendix): each label's distinctiveness is essentially invariant to K over the range tested, with the sole exception of one theme whose dominant factor only resolves at K of fifteen or above, so K = 15 is the smallest count that resolves all themes at no cost to the others.

3.4 Theme Fingerprints and Projection

Each theme's ETF returns are residualized identically and regressed on the K extracted factors, yielding a K-dimensional loading vector — the theme's fingerprint in factor space. Per theme, the fingerprint is the centroid of its constituent ETFs' loading vectors, which averages out individual ETF construction idiosyncrasies. Every stock in the target universe is then projected into the same factor space, and scored by cosine similarity between its loading vector and each theme fingerprint. The top-ranked names per theme form the candidate basket.

3.5 Diagnostics: Coherence and Distinctiveness

Two diagnostics, computed before basket construction, determine whether a theme is recoverable. Coherence measures how concentrated a theme's fingerprint is across factors, computed as the average of three normalized concentration metrics (a concentration ratio, a Herfindahl index, and an entropy-based effective-factor count). Distinctiveness measures whether a theme uniquely owns its dominant factor or shares it with other themes. The accompanying factor-sharing map records, for each factor, which themes load on it most strongly. As shown in Section 4, distinctiveness rather than coherence is the operative criterion for clean baskets.

3.6 Validation

Candidate baskets are validated out-of-sample against the theme ETFs along two axes. The first is correlation: whether the equal-weighted candidate basket co-moves with the theme ETF. The second is a beta decomposition: regressing the basket return on the ETF return to separate co-movement (correlation) from amplitude (beta), and recovering an annualized alpha. This two-axis validation distinguishes baskets that track the theme at full amplitude from those that are muted, and isolates the effect of mega-cap exclusion from genuine signal attenuation.

3.7 Data and Proxies

Several inputs are proxied owing to data-access constraints; each is noted for reproducibility. MSCI ACWI constituents require a vendor license and are proxied by free index lists, ultimately restricted to US listings. ETF constituent holdings are not used at all — ETF price series serve as theme labels, which is the method's holdings-free feature rather than a limitation. The Russell 2000 constituent list requires an FTSE Russell license and is proxied by a metadata-filtered NASDAQ and NYSE cross-section. There is no canonical commodity factor library; we proxy the Bakshi-Gao-Rossi commodity factors with liquid commodity-sector ETFs.

4. Results

We organize the results as a sequence of findings, each building on the previous. We report results for eleven cross-sector themes — Robotics & AI, AI Infrastructure, Clean Energy, Cybersecurity, Defense, Fintech, US Infrastructure, Agribusiness, Critical Minerals, Water, and Timber & Forestry — together with six calibration controls: three GICS sectors (Financials, Technology, Energy) and three subsectors (Banks, Software, Biotech). Themes are admitted only if their defining ETF ensemble spans multiple GICS sectors; single-industry funds such as biotech ETFs are excluded as themes and repurposed as subsector controls. This admissibility rule is applied as an ex-ante filter, not a post-hoc selection.

4.1 Themes Are Weak but Recoverable Latent Factors

After residualization and standardization, every extracted factor corresponds to an interpretable sector or style axis (for example, utilities versus semiconductors, banks versus energy, REITs versus homebuilders). Themes appear as latent factors with best-single-factor R-squared values in the range of approximately 0.08 to 0.21. These are weak factors by construction — they explain a modest share of even the residual variance — but they are clearly present and economically interpretable, not statistical artifacts.

4.2 Clean Factors Require a Clean Universe

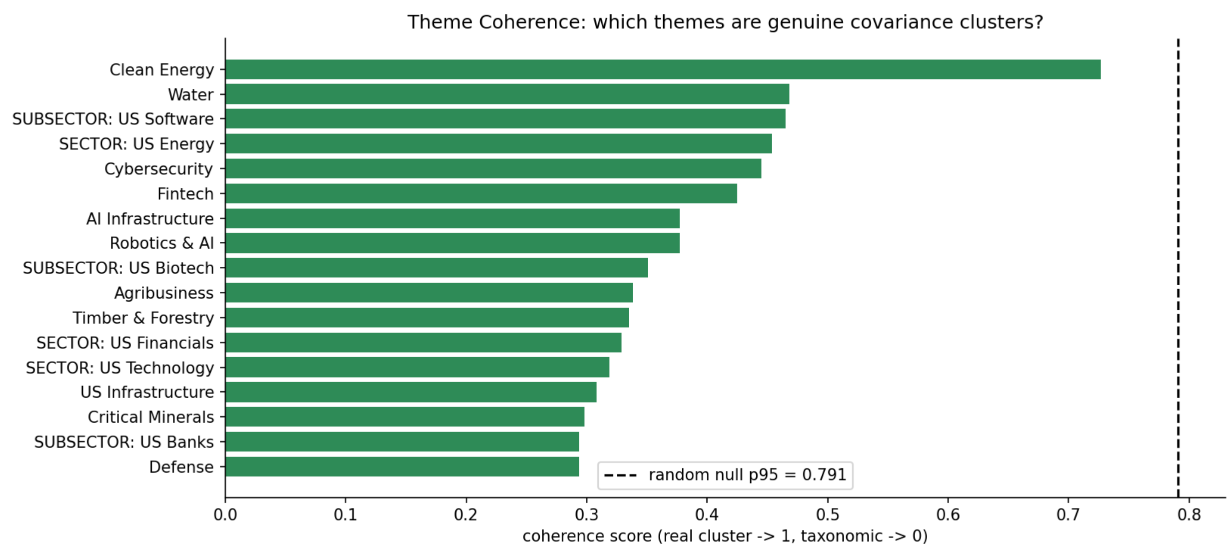
This finding is methodological but consequential. Before imposing the market-capitalization floor and standardization, extracted factors were dominated by individual volatile micro-caps — in one instance a single stock carried a loading of 0.89 — and by penny and cryptocurrency-linked names. Two interventions were decisive: a USD 1 billion floor on the covariance universe, and winsorization plus unit-volatility standardization of residuals. After both, the factor structure became cleanly interpretable. Statistical theme detection is only as reliable as the cleanliness of the covariance universe on which factors are estimated.

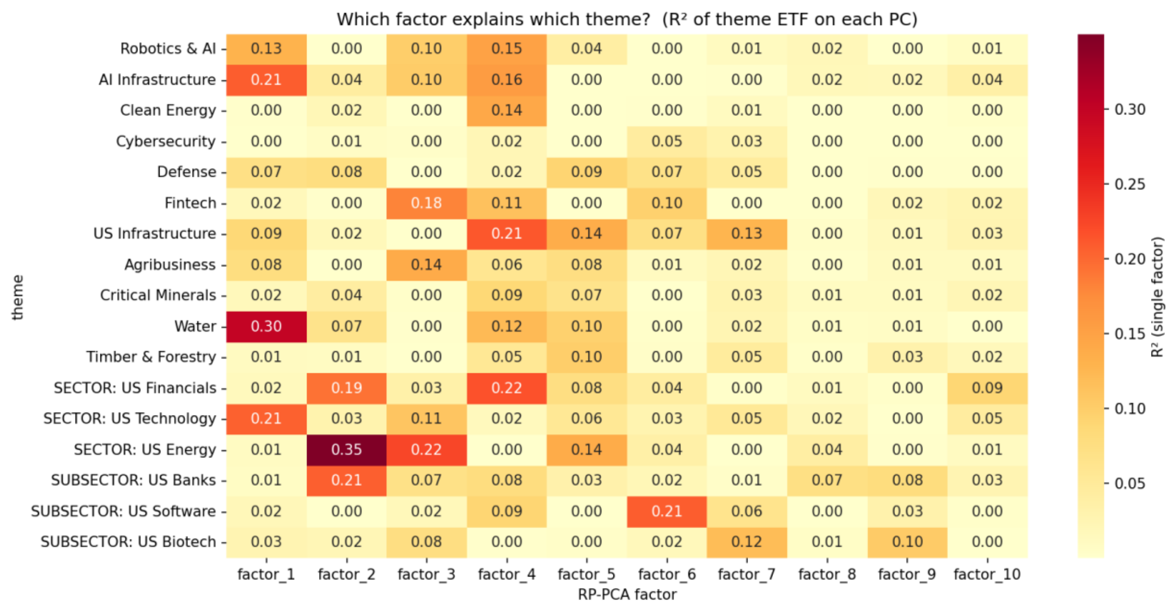
4.3 A Theme Is Tradeable If and Only If It Uniquely Owns a Factor

This is the central finding. Coherence alone does not predict basket quality — a theme can be highly concentrated yet produce contaminated baskets if its dominant factor is shared. The operative criterion is distinctiveness: a theme produces clean, correct baskets when it uniquely owns a distinct latent factor, and suffers mutual contamination when several themes load on the same factor. The factor-sharing map below, from the baseline specification, makes the mechanism explicit.

Factor	Themes loading on it (baseline)
factor_1	AI Infrastructure (unique)
factor_3	Fintech, Agribusiness (shared)
factor_4	US Infrastructure, Robotics & AI, Clean Energy, Critical Minerals (shared)
factor_5	Defense (unique)
factor_6	Cybersecurity (unique)

Table 1. Factor-sharing map, baseline specification (FF5 residualization, no commodity controls). Themes that uniquely own a factor (AI Infrastructure, Defense, Cybersecurity) yield clean baskets; the four themes sharing factor_4 mutually contaminate one another.





The themes that uniquely own a factor — AI Infrastructure, Defense, and Cybersecurity — produce the cleanest baskets. The four themes piling onto factor_4 (a broad cyclical and materials axis) bleed into one another: Clean Energy baskets fill with mining names because Critical Minerals shares its factor.

4.4 Commodity-Linked Themes Fuse via Shared Commodity Beta

The factor_4 pileup has an economic explanation: clean energy and critical-minerals companies share substantial commodity-input and capital-cycle exposure, and absent an explicit control, that shared commodity beta forms a single latent factor onto which multiple themes load. Adding commodity-sector ETF controls to the residualization (proxying the Bakshi-Gao-Rossi factors) breaks the pileup apart. After commodity residualization, the four-theme factor_4 cluster separates, and the affected baskets clean up materially.

Theme	Baseline basket character	With commodity controls
Clean Energy	Gold miners (AU, GFI, AEM, KGC, WPM, FNV)	Genuine clean-energy (BELFA, ENS, WATT, ASPN, CSIQ, TAC)
Agribusiness	Pipelines and miners (ENB, FNV, KGC)	Food and agriculture (POST, PPC, BJ, GEF)

Table 2. Effect of commodity-factor controls on two contaminated themes. Clean Energy ETF correlation rose from 0.70 to 0.88 after commodity residualization.

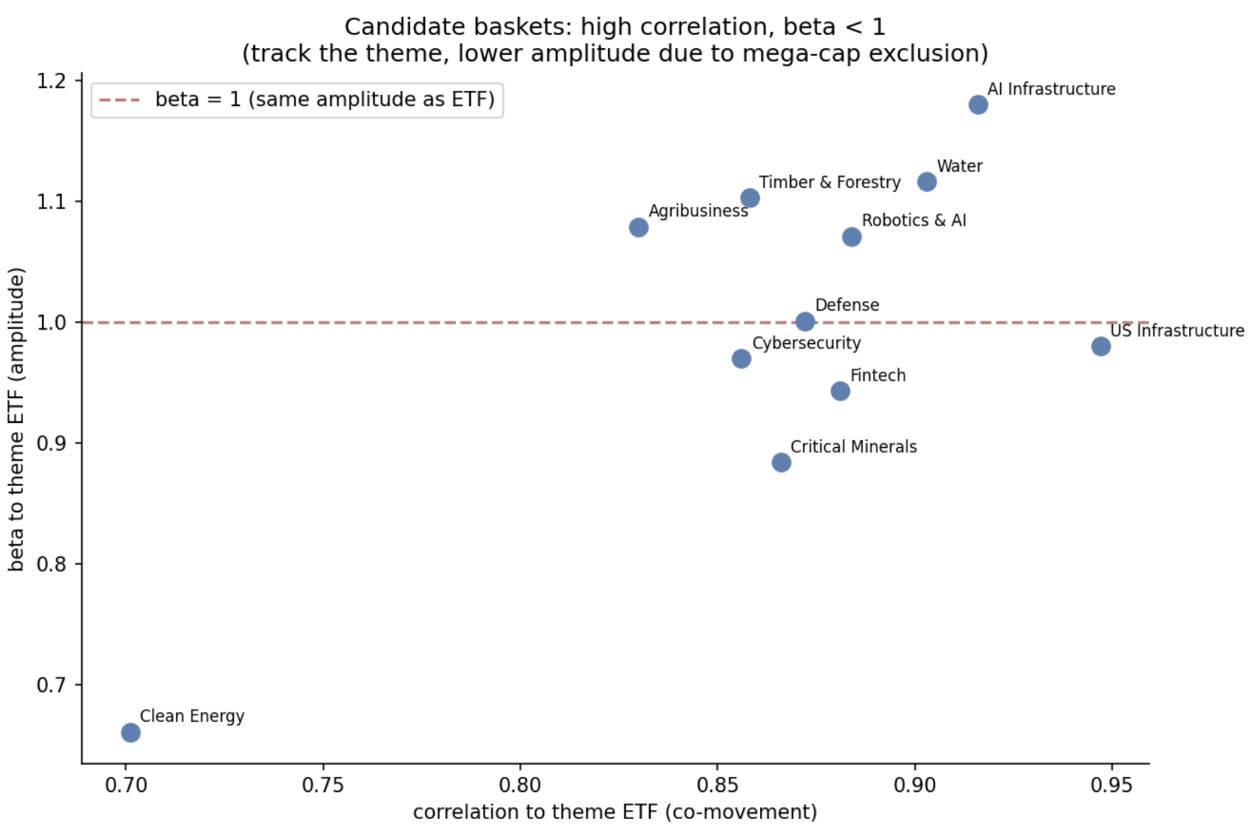
This is both a limitation and a prescription: themes whose narrative driver coincides with a commodity factor require explicit commodity controls to separate the thematic signal from shared commodity beta.

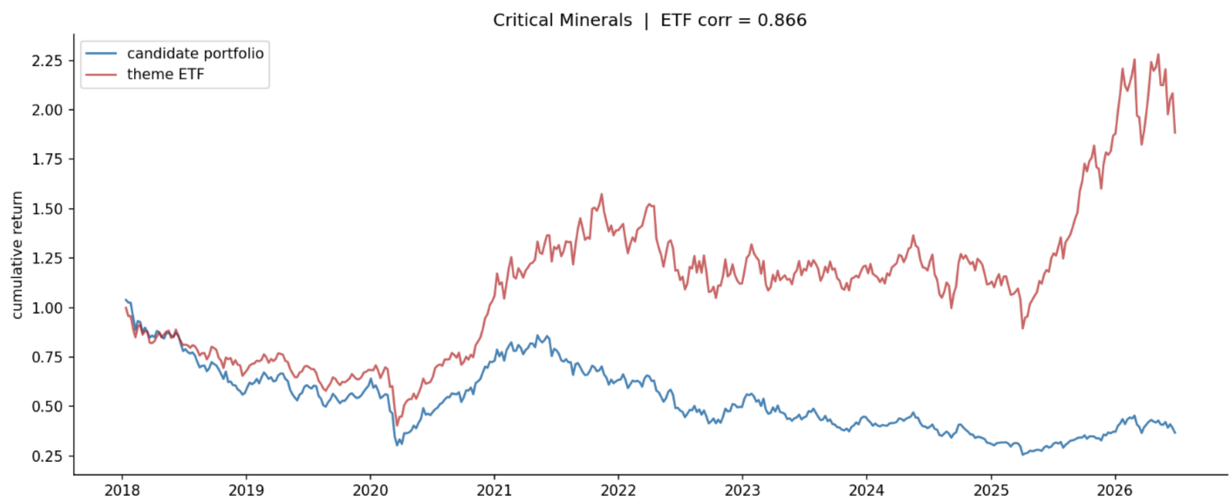
4.5 Validation: Two-Regime Beta

Out-of-sample validation against theme ETFs reveals high correlation across all themes (0.78 to 0.95), confirming that candidate baskets co-move with their themes. The beta decomposition adds a sharper result: baskets fall into two amplitude regimes that map onto the factor-ownership finding.

Theme	ETF corr.	Beta	Basket cum.	ETF cum.
AI Infrastructure	0.92	1.18	2.13	4.69
Robotics & AI	0.88	1.07	1.18	0.65
Defense	0.87	1.00	1.39	1.36
US Infrastructure	0.94	0.98	1.79	1.49
Agribusiness	0.78	1.08	0.01	0.40
Fintech	0.88	0.94	-0.31	-0.04
Cybersecurity	0.83	0.97	-0.11	1.59
Critical Minerals	0.81	0.89	-0.63	0.92
Clean Energy	0.70	0.66	-0.35	0.40

Table 3. Out-of-sample validation. Correlation is uniformly high; beta separates themes that track at full amplitude (beta >= 1) from commodity-contaminated themes (beta < 1). Cumulative returns are over the 2018-2026 sample.




Critical Minerals | ETF corr = 0.866
candidate portfolio
theme ETF
cumulative return
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
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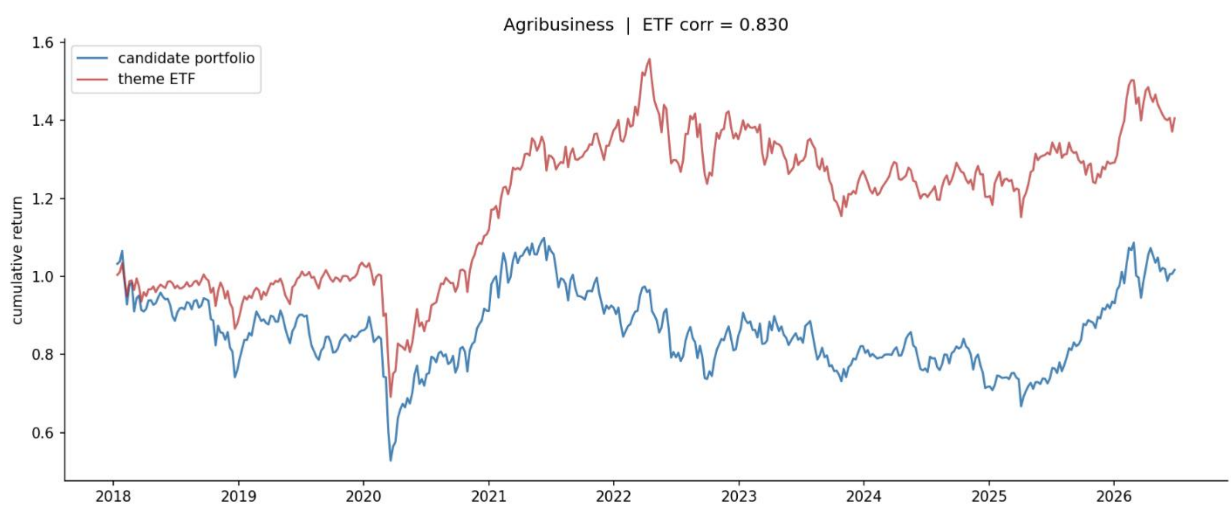
Agribusiness | ETF corr = 0.830
candidate portfolio
theme ETF
cumulative return
1.6
1.4
1.2
1.0
0.8
0.6
2018
2019
2020
2021
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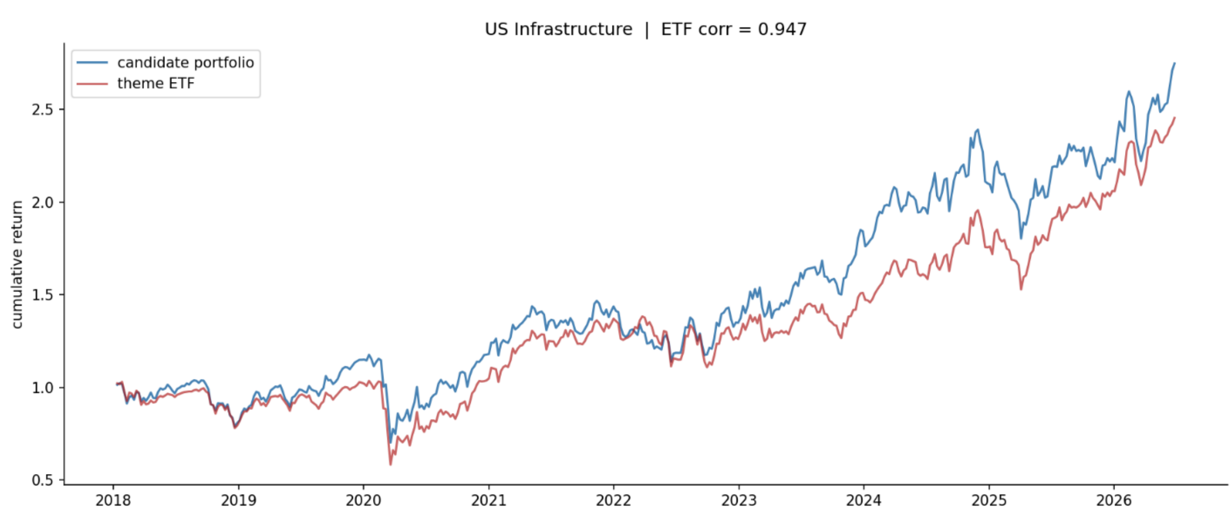
2026


US Infrastructure | ETF corr = 0.947
candidate portfolio
theme ETF
cumulative return
2.5
2.0
1.5
1.0
0.5
2018
2019
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2021
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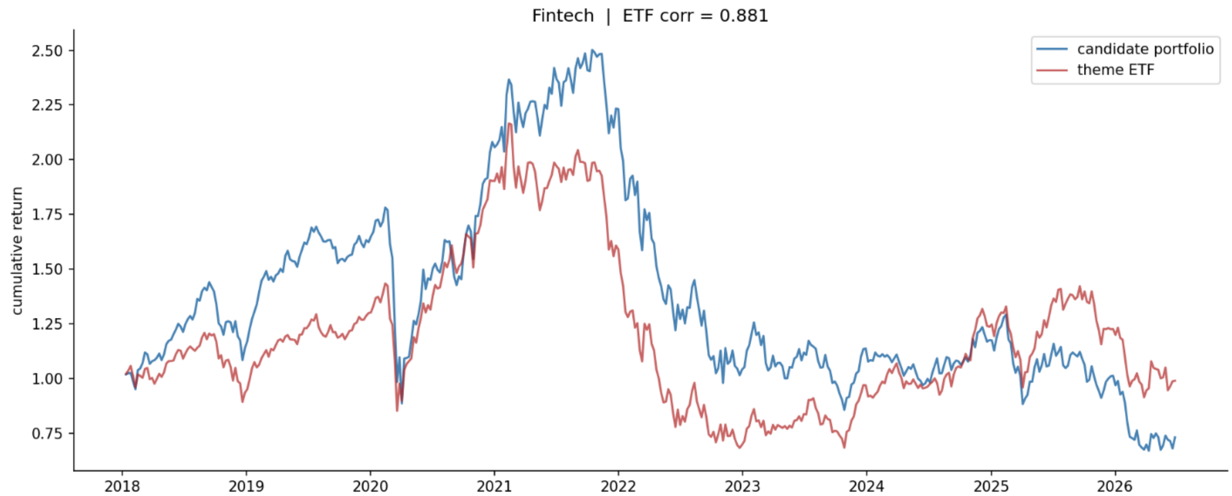
Fintech | ETF corr = 0.881
candidate portfolio
theme ETF
cumulative return
2.50
2.25
2.00
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0.75
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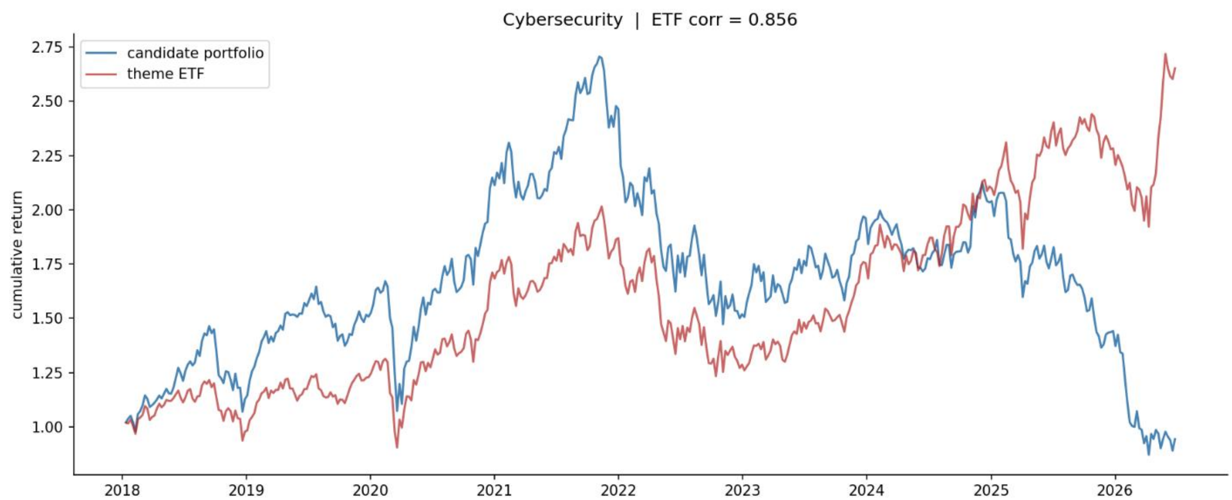
Cybersecurity | ETF corr = 0.856
candidate portfolio
theme ETF
cumulative return
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1.00
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2024
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2026


Defense | ETF corr = 0.872
candidate portfolio
theme ETF
cumulative return
2.5
2.0
1.5
1.0
0.5
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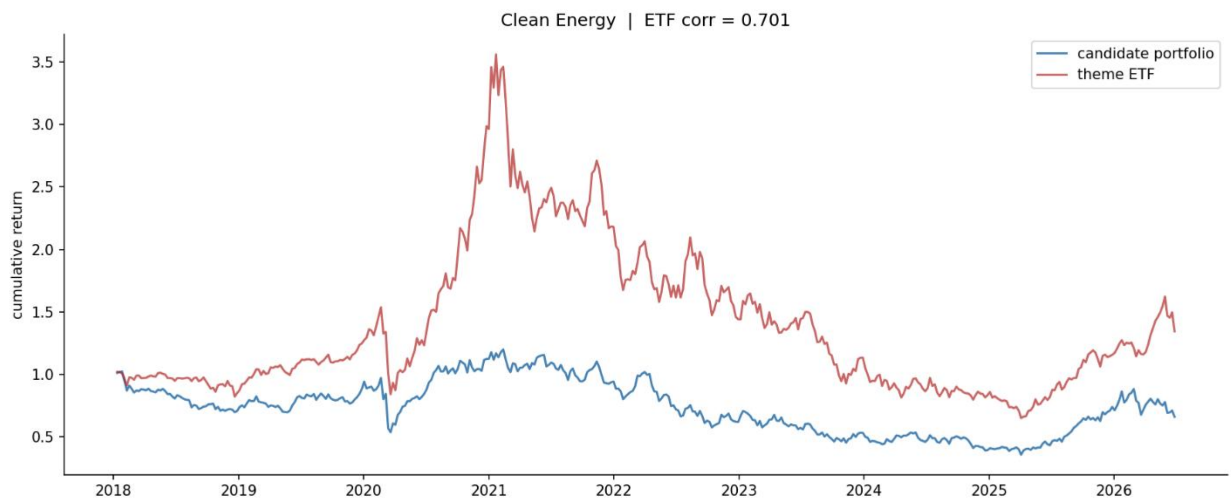
Clean Energy | ETF corr = 0.701
candidate portfolio
theme ETF
cumulative return
3.5
3.0
2.5
2.0
1.5
1.0
0.5
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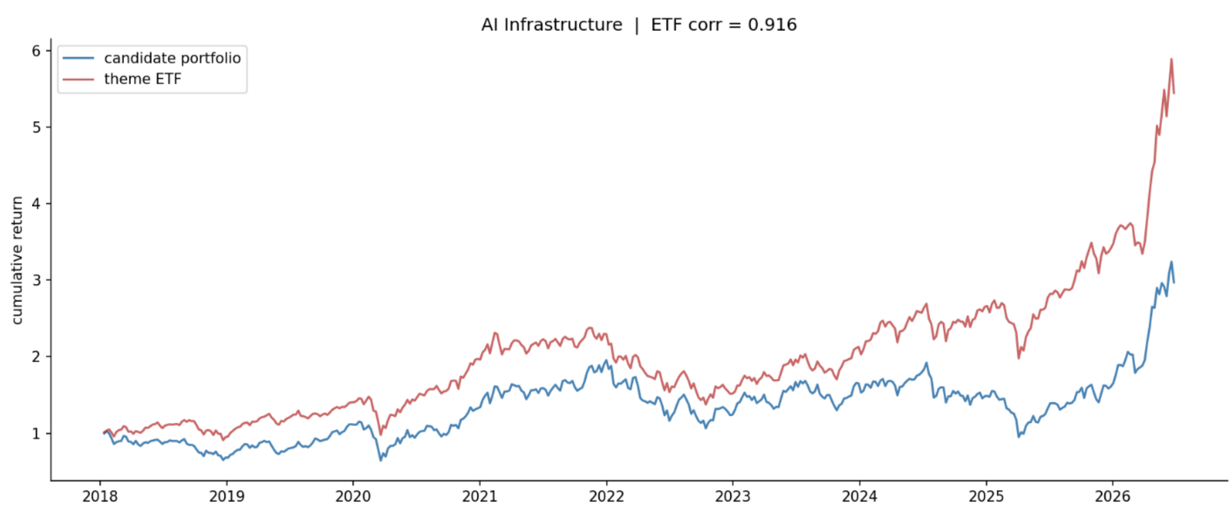
AI Infrastructure | ETF corr = 0.916
candidate portfolio
theme ETF
cumulative return
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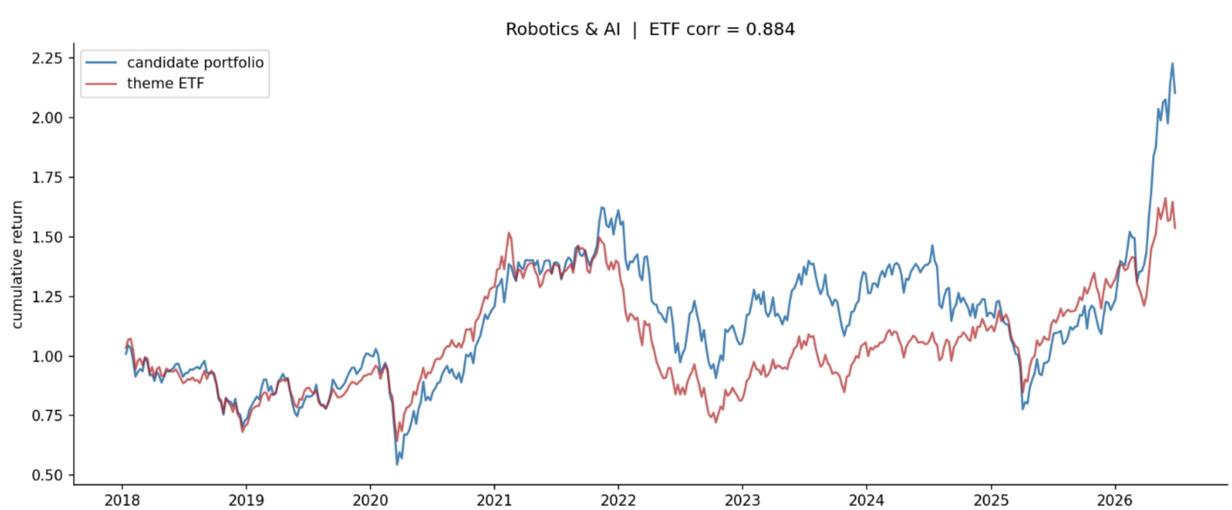
Robotics & AI | ETF corr = 0.884
candidate portfolio
theme ETF
cumulative return
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
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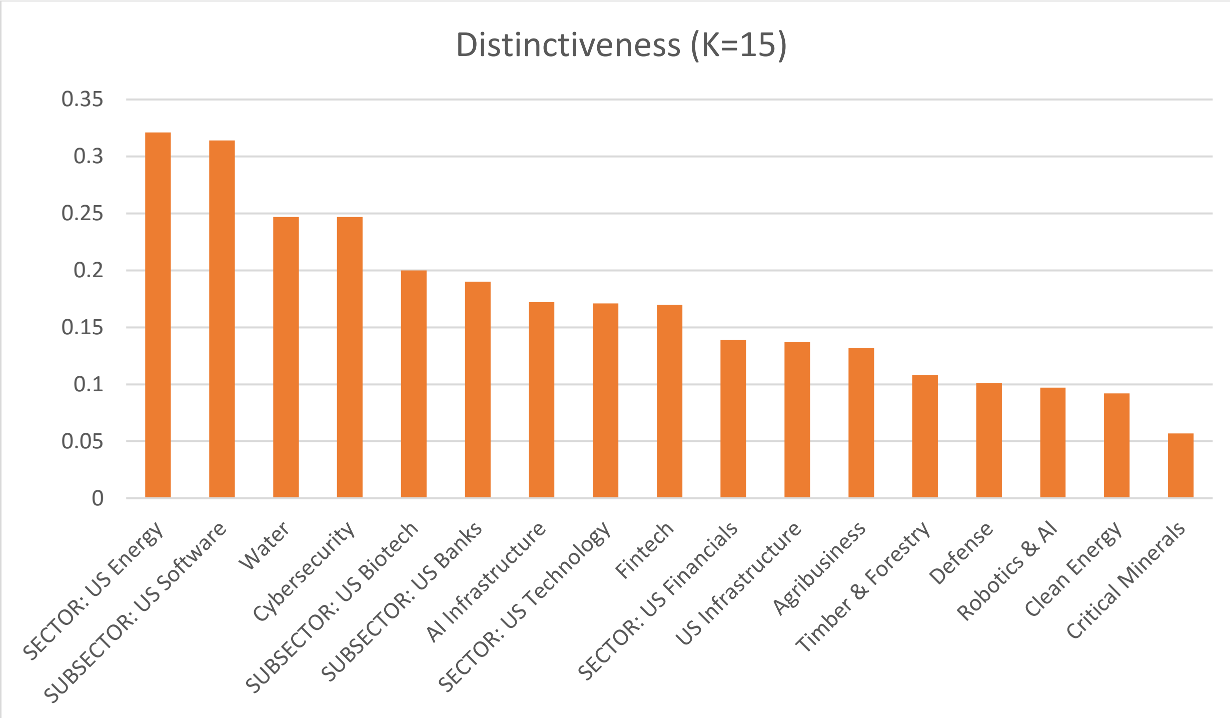
Themes that own a clean factor track their ETF at full amplitude (beta at or above 1): AI Infrastructure 1.18, Robotics & AI 1.07, Defense 1.00, US Infrastructure 0.98, Agribusiness 1.08. Themes that are commodity-contaminated are muted (beta below 1), most starkly Clean Energy at 0.66. Where a level gap coexists with beta at or above 1 — AI Infrastructure tracks at beta 1.18 yet reaches only 2.13 against the ETF's 4.69 — the gap is driven by mega-cap exclusion: the small-cap basket co-moves at full amplitude with the theme factor but structurally cannot hold the cap-weighted mega-cap winners that drive the ETF's level.

4.6 Coherence Is Regime-Dependent, Not Taxonomic (*or Validating the Diagnostic Against GICS Controls*)

To calibrate the coherence scale, we introduced GICS sectors and subsectors as control labels, expecting them to sit at the top — a sector should own a latent factor almost by construction. The data contradict this expectation, and the contradiction is the paper's most consequential empirical result. Ranking all labels by distinctiveness, the GICS sectors are scattered through the themes rather than above them. US Financials, a textbook sector, ranks near the bottom, below cross-sector themes such as Water and AI Infrastructure.

Label	Type	Distinctiveness	Dominant factor
US Energy	sector	0.319	factor_2
US Software	subsector	0.314	factor_6
Water	theme	0.247	factor_1
US Biotech	subsector	0.200	factor_7
US Banks	subsector	0.189	factor_2
AI Infrastructure	theme	0.172	factor_1
US Technology	sector	0.171	factor_1
Fintech	theme	0.170	factor_3
US Financials	sector	0.139	factor_4
US Infrastructure	theme	0.137	factor_4

Table 5. Distinctiveness of themes and GICS controls (top ten of seventeen labels). Sectors are interleaved with themes rather than dominating; US Financials ranks below the cross-sector themes Water and AI Infrastructure.



The mechanism is visible in the factor-sharing structure. The broad Technology sector loads on one factor, while its Software subsector loads on a different factor and semiconductors load on a third. The sector is split across factors because its subindustries have decoupled. In the 2018-2026 sample, the dominant structural driver — capital expenditure on artificial intelligence — has separated semiconductors and AI-infrastructure names from the rest of technology, so the broad sector is a less coherent cluster than its own fragments. The same fracture explains why US Financials, spanning banks, insurers, payments, and capital markets that no longer move as one, fails to own a factor.

This yields a reinterpretation of coherence itself. Coherence is not a fixed attribute of taxonomy — theme versus sector — but a regime-dependent attribute of co-movement. A GICS sector is a coherent covariance cluster only when its subindustries co-move; when a structural force splits a sector internally, the subindustry-spanning theme becomes the better-defined cluster. A theme, on this view, is a grouping that has decoupled from its GICS parent because a new economic driver cuts across the old classification. A theme is economically real precisely when it is more coherent than the sectors it draws from — and Water outscoring US Financials is the proof of concept, not an anomaly. We confirm the ordering is invariant to the number of extracted factors over the range tested (Appendix), so it is a stable structural property of the regime rather than an artifact of factor count.

5. Discussion

5.1 Purity versus Amplitude

A natural objection is that small-cap baskets underperform their large-cap ETFs in level terms. This does not indicate failure; it reflects a distinction between purity and amplitude. Small-cap names typically have higher thematic purity — a pure-play small-cap derives a larger share of its economics from the theme than a diversified large-cap. But an equal-weighted small-cap basket can exhibit lower realized amplitude than a cap-weighted ETF, because the ETF concentrates in the cap-weighted mega-cap winners while the equal-weighted basket carries the full cross-section, including the idiosyncratic losers that characterize the fat left tail of small-cap thematic universes. Purity is a per-name property about exposure; amplitude is a portfolio-level property about realized return. This motivates more sophisticated basket construction — beta-weighting and idiosyncratic-risk trimming — as future work.

5.2 The Diagnostic as a Pre-Trade Filter

The coherence and distinctiveness diagnostics are valuable independently of the discovery engine. They constitute a test of whether a theme is economically real — a genuine covariance cluster that the market prices as a unit — or merely a taxonomic label imposed on companies that belong to different covariance clusters. A theme that does not resolve into a distinct factor is telling the practitioner that the market does not trade it as a coherent unit, which is itself decision-relevant for a thematic portfolio manager assessing whether a theme is tradeable.

5.3 Practical Use

For a multi-strategy portfolio manager with a days-to-weeks horizon, the value of the engine is not in replicating an ETF's return — that exposure is directly purchasable — but in surfacing the small-cap expression of a theme, which is typically less crowded and capacity-constrained than the mega-cap names already crowded by large funds. A basket that reliably co-moves with a theme at correlation near 0.9 is a tradeable instrument: it can be sized, hedged against the ETF or the crowded large-caps, and rotated. The amplitude gap is the opportunity set, not the error term.

5.4 The Momentum Question: Factor Ownership Predicts Independence

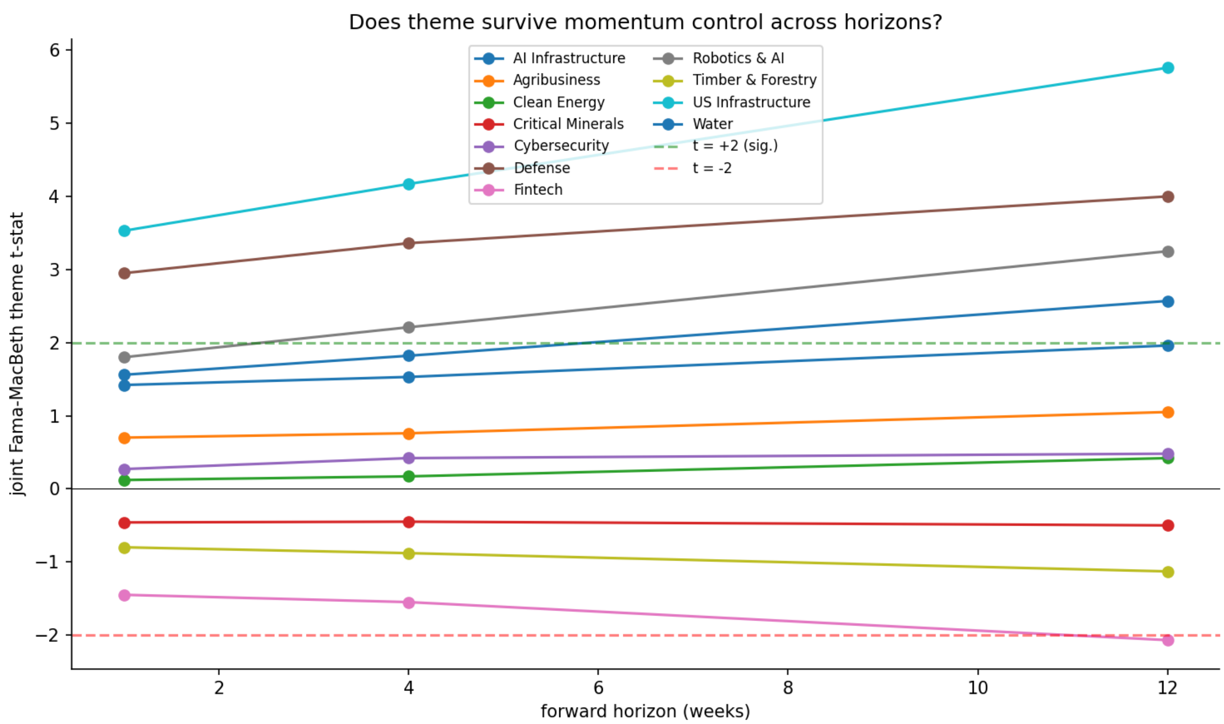
A recurring industry critique holds that thematic investing is repackaged momentum. We test this directly with a three-part battery applied to the discovered small-cap candidates: an independent double sort (does the theme-score return spread survive within momentum quintiles), Fama-MacBeth cross-sectional regressions (does the theme coefficient survive controlling for each stock's own momentum, with Newey-West standard errors), and a spanning regression (does the theme long-short portfolio earn alpha relative to the momentum factor and the Fama-French five). Throughout, momentum is each stock's own trailing twelve-minus-one-month return computed in the target universe — a distinct object from the momentum factor used in residualization.

The answer is not universal, and that is the finding. It is theme-specific, and it lines up precisely with the factor-ownership diagnostic of Section 4.3. The themes that uniquely own a latent factor are exactly the themes whose signal survives the momentum control.

Theme	Owns factor?	Double sort	FM joint t	Spanning alpha
US Infrastructure	strong	+1.31%	4.19	+16.3% (t 3.77)
Defense	unique	+1.22%	3.37	+12.8% (t 3.06)

Robotics & AI	shared	+0.61%	2.24	n.s.
AI Infrastructure	unique	+0.58%	1.84	n.s.
Agribusiness	weak	+0.21%	0.77	n.s.
Cybersecurity	weak	+0.18%	0.40	n.s.
Clean Energy	contaminated	+0.02%	0.18	n.s.
Critical Minerals	shared	-0.26%	-0.44	-8.8% (n.s.)
Fintech	shared	-0.57%	-1.57	-8.5% (n.s.)

Table 4. Momentum battery on discovered candidates, ordered by double-sort spread. Themes that uniquely own a factor (US Infrastructure, Defense) survive all three tests; themes that share a factor or own none show no independent signal, and two show negative signal. Double sort is the theme Q5-minus-Q1 forward-return spread averaged within momentum quintiles; FM joint t is the Newey-West t-statistic on the theme coefficient after adding momentum.



Two themes — US Infrastructure and Defense — pass all three tests decisively, with theme coefficients essentially unchanged by the addition of momentum. These are proof cases that the discovered signal is not momentum in disguise. The themes that share a factor or own none show no significant independent signal, and Fintech and Critical Minerals show negative signal, indicating their candidate baskets are on the wrong side of momentum. Crucially, it is not random which themes pass: the survivors are precisely the factor-owners identified in advance by the coherence diagnostic, which both defuses the multiple-testing concern and integrates the momentum question into the paper's central mechanism. The honest answer to whether thematic investing is just momentum is therefore conditional — it depends on whether the theme is a coherent, distinctly-owned factor, and the coherence diagnostic predicts which themes those are.

We confirm these results across forward-return horizons of one, four, and twelve weeks. US Infrastructure and Defense remain significant at every horizon (robust three-of-three), while the null themes remain flat near zero throughout. Notably, for the two robust themes the t-statistic rises monotonically with horizon — for US Infrastructure from 3.53 at one week to 5.76 at twelve weeks, and for Defense from 2.95 to 4.00. A genuine momentum echo would decay as the horizon lengthens; the fact that the theme signal instead strengthens indicates it is a slow-moving structural signal that compounds over weeks, providing additional mechanistic separation between theme and momentum.

5.5 Limitations

- The out-of-sample validation metric, ETF correlation, is necessary but not sufficient: a basket may correlate with an ETF for non-thematic reasons. A second validation axis based on fundamental sector or industry overlap would strengthen the claims.
- The number of factors K is set to 15 on the basis of a sweep showing distinctiveness is invariant to K across the tested range (Appendix); formal information-criterion selection is left to future work.
- The momentum battery uses each stock's own trailing return as the momentum control; results are reported across one-, four-, and twelve-week horizons, but a longer sample and additional momentum specifications would further strengthen the conditional claim.
- Several data inputs are proxied owing to licensing constraints, which may introduce off-theme holdings and attenuate measured performance.

6. Conclusion

We set out to recover investment themes from return co-movement alone, to determine what makes a theme tradeable, and to test whether thematic exposure is distinct from momentum. ThemeCloner extracts a theme's signature from the residual co-movement of thematic ETFs and projects it onto a disjoint small-cap universe, discovering thematically-exposed stocks without using any holdings data. The method's central output is not a single basket but a diagnostic: a theme yields clean, investable baskets if and only if it uniquely owns a latent factor, and this same criterion of factor ownership predicts which themes contaminate one another and which survive a momentum control.

The most consequential finding is that coherence is regime-dependent rather than taxonomic. Introduced as a calibration ceiling, GICS sectors instead scatter among the themes, and several cross-sector themes prove to be more distinctly-owned latent factors than the sectors they span. The reason is that a structural driver — artificial-intelligence capital expenditure in our sample — fractures sectors internally, so that the subindustry-spanning theme becomes the better-defined cluster. This supplies an economic rationale for thematic investing that does not rest on narrative: themes matter because sectors cohere only when their subindustries do, and when a new driver cuts across the old taxonomy, the theme is the more faithful description of how the market actually moves.

On the momentum question, our answer is conditional and tied to the same mechanism. The discovered signal survives controls for stock-level momentum, across forward horizons, precisely for the themes that own a distinct factor; for themes that share a factor or own none, the signal is weak or absent. Thematic investing is therefore not reducible to momentum, but neither is every theme distinct from it — the distinction holds exactly when the theme is a genuine, distinctly-owned covariance cluster. The coherence diagnostic, computed before any basket is formed, tells the practitioner in advance which case they are in.

Several extensions follow naturally. The regime-dependence result invites a multi-period study comparing how the coherence ordering shifts across historical regimes, which our framework is constructed to run. The validation can be strengthened with a fundamentals-based axis beyond return correlation, and the small-cap amplitude gap invites more sophisticated basket construction. We leave these to future work, having established within the present regime that themes are recoverable latent factors, that factor ownership determines tradeability, and that the long-standing question of whether thematic investing is merely momentum has a precise, mechanism-based answer.

Appendix A. Robustness of Coherence to the Number of Factors

The coherence and distinctiveness diagnostics depend on the number of extracted factors, K. With seventeen labels and a finite factor count, factor-sharing is partly mechanical: when there are fewer factors than distinct clusters, labels are forced to share. To confirm that our findings — in particular the regime-coherence ordering in which cross-sector themes out-rank GICS sectors — are not artifacts of a particular K, we re-estimate the factor model and recompute distinctiveness at K of 10, 15, 20, and 25.

The ordering is essentially invariant to K. Each label's distinctiveness is stable across the tested range, with the single exception of Cybersecurity, whose dominant factor only resolves at K of fifteen or above. We therefore adopt K = 15 as the default in the main analysis: it is the smallest factor count that resolves every theme, and it leaves all other labels' distinctiveness unchanged. Critically, the headline comparison is robust at every K: the cross-sector theme Water is more distinctly-owned than the US Financials sector at K of 10, 15, 20, and 25 alike.

Label	Type	K=10	K=15	K=20	K=25
US Energy	sector	0.321	0.321	0.321	0.321
US Software	subsector	0.314	0.314	0.314	0.314
Water	theme	0.247	0.247	0.247	0.247
Cybersecurity	theme	0.068	0.247	0.247	0.247
US Biotech	subsector	0.200	0.200	0.200	0.200
US Banks	subsector	0.190	0.190	0.190	0.190
AI Infrastructure	theme	0.172	0.172	0.172	0.172
US Technology	sector	0.171	0.171	0.171	0.171
Fintech	theme	0.170	0.170	0.170	0.170
US Financials	sector	0.139	0.139	0.139	0.139

US Infrastructure	theme	0.137	0.137	0.137	0.137
Agribusiness	theme	0.132	0.132	0.132	0.132
Timber & Forestry	theme	0.108	0.108	0.108	0.108
Defense	theme	0.101	0.101	0.101	0.101
Robotics & AI	theme	0.097	0.097	0.097	0.097
Clean Energy	theme	0.092	0.092	0.092	0.092
Critical Minerals	theme	0.057	0.057	0.057	0.057

Table A1. Distinctiveness by number of factors K, all seventeen labels, sorted by mean across K. Values are invariant to K except Cybersecurity (resolves at K>=15). Water (theme) exceeds US Financials (sector) at every K.